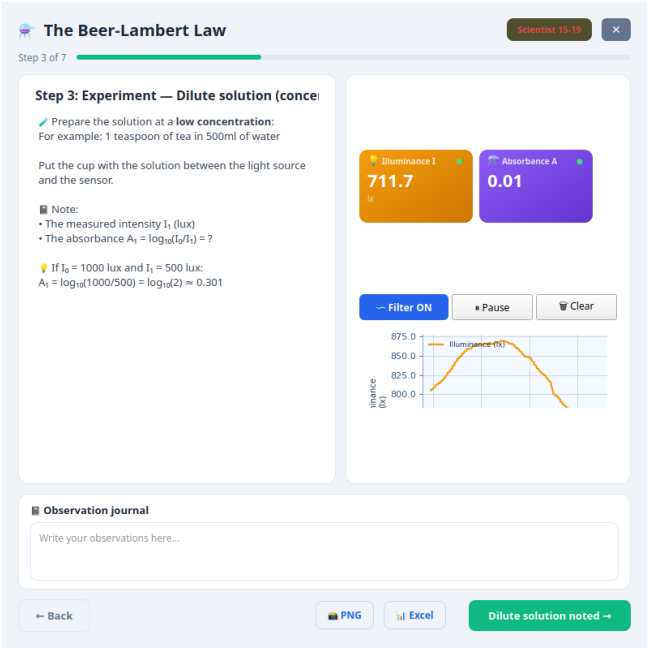
The Beer–Lambert Law

A dilution series, a logarithm, and the precise point where the law breaks

Level: Scientist — ages 16–19 (Romanian school years X–XII) • **Time in class:** 50 min, or 3 lessons of 50 min • **Sensors:** The light sensor (illuminance)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Scientist — **ages 16–19 (Romanian school years X–XII)**, and the lower boundary is strict: the whole activity *is* a logarithm, and **logarithms enter the Romanian mathematics syllabus in year X**. Unlike *Spectroscopy*, there is no honest transmittance-only version here — without the logarithm there is neither absorbance nor a straight line. The most direct entries are in **year X** (mathematics: logarithms and the logarithmic function), and **year IX** (chemistry: solutions and concentration, revisited here). **The Beer–Lambert law is not in the core syllabus**: it is enrichment, science club and olympiad content.

Assumed knowledge

The base-ten logarithm and its properties. Preparing a dilution with a syringe and computing relative concentration. An XY chart with a line of best fit in a spreadsheet; for the extension, R^2.

Time in class

Short: 50 min, with the solutions prepared by you in advance. Full: three 50-minute lessons — preparing the series, measuring, analysing. About two and a half hours of class time in total, plus 30–40 minutes of preparation.

Teacher preparation

30–40 min — make one stock solution for the whole class and **run the experiment yourself once**, checking a single thing: that the stock still transmits a usable amount of light. **If it reads below about five per cent of the blank, dilute it** — a solution that blocks nearly all the light gives an absorbance that is pure noise. Aim for a stock absorbance between 0.8 and 1.2. Then decant into labelled bottles, one per group.

Sensor settling time	**5 seconds** after each change of sample — the light sensor responds practically instantly. **The real wait is the lamp's**: switch it on ten minutes before the lesson, because a cold lamp changes its output over the first few minutes and I_0 is captured only once.
Sensors and boxes	The light sensor — the only one required. It measures broadband visible illuminance, not a single wavelength, and section 4 explains what follows from that. The activity header shows the box in use; if several are connected, choose the group's box in the header.

Three routes

Short	50 min: predictions (5), theory (7), calibration on the blank (5), the six pre-prepared samples (15), the app's graph and one absorbance worked by hand (8), the blank again — "How sure are we?" (3), conclusions (7). Pupils do not dilute.
Full	Three 50-minute lessons: lesson 1 predictions, theory, preparing the dilution series and building the light tunnel; lesson 2 measurement, blank first, then each concentration, plus a deliberately over-concentrated sample; lesson 3 the analysis, the line, the gradient, R^2 and the discussion of deviations.
Project	Extension: the same series measured through a complementary coloured filter, to see the top of the line straighten; or varying the path length instead of the concentration.

The rig matters more than the chemistry

Almost every failed run in this activity is a **geometry failure, not a chemistry failure**. If the lamp, the cup and the sensor are not in exactly the same positions for every reading, the comparison means nothing.

Spend the first ten minutes building a fixed rig — a cardboard box lying on its side with the lamp taped at one end, the sensor at the other, and a marked rectangle on the floor of the box where the cup always sits — and the rest of the activity works. Mark the positions with tape and forbid anyone to move them.

1. Where this fits in the curriculum

- **Mathematics** — in **year X**, logarithms and the logarithmic function. This is the central and most valuable entry: pupils do not apply a given formula, they see *why* somebody invented absorbance. The raw data in lux fall on a curve; the same data, logged, fall on a straight line. **The logarithm appears here as a tool for straightening curves**, which is exactly its historical role, rather than as a chapter in a textbook.
- **Chemistry** — in **year IX**, solutions and concentration, revisited as a dilution series. Colorimetry and the determination of an unknown concentration from a calibration line — the real analytical method — are **not in the core syllabus**: they are enrichment content, for a chemistry club, olympiad preparation and specialist classes. Check the syllabus in force for your year and stream.
- **Physics** — absorption of light in a medium, the exponential law of attenuation, and the difference between absorption and scattering; again as enrichment rather than a compulsory topic.
- **Biology and applications** — pulse oximetry, water turbidity, quality control in the food industry: all use exactly this law, and the app's final step lists them.
- **Computing and ICT** — processing their own series: the transmittance and absorbance columns computed with formulae, the XY chart, the line of best fit and R^2.

The core skill is **linearisation**: taking a curved relationship, applying the transformation that straightens it, and knowing *where* the straightening fails. That last part matters most — which is

why the series contains a deliberately over-concentrated sample.

2. Time plan

The in-app wizard has nine steps and runs as a single session: a written prediction before measuring, calibration on the blank, six samples over three measuring steps, the graph, the blank check and the conclusion. The version worth teaching uses a five- or six-point series and gives the analysis the time it deserves, which means three lessons plus your own preparation the day before.

Route	Time	What happens
Short	**50 min**	Predictions (5), theory (7), calibration on the blank (5), the six pre-prepared samples over the three measuring steps (15), the app's graph and one absorbance worked by hand (8), the blank again at the "How sure are we?" step (3), conclusions (7). Use solutions you made yourself; pupils do not dilute.
Lesson 1	50 min	Written predictions; theory and the derivation of $A = \varepsilon c l$; pupils prepare their own dilution series by syringe from the stock; building and aligning the light tunnel.
Lesson 2	50 min	Measurement: **the blank first**, to fix I_0, then each concentration in turn, at least three readings each. A deliberately over-concentrated sample is measured last, so that the breakdown of the law can be seen. Data exported at the end.
Lesson 3	50 min	Analysis and write-up: the transmittance and absorbance columns, the A against c graph, the gradient, R^2, the discussion of deviations, and the path-length extension for fast groups.

In total the full version costs **about two and a half hours of class time across three days**, plus **thirty to forty minutes of teacher preparation** the afternoon before, for the stock solution and a trial run. The short version is a single lesson, but only if the solutions are made in advance — the dilution work alone occupies twenty minutes.

3. Materials

Item	Per group	Notes
MagicBox with the light sensor	1	The only sensor required. It measures broadband visible illuminance, not a single wavelength — see the background section for why that matters.
Laptop with the app	1	Kept well away from the water and the solutions. The box is not waterproof either.
Identical transparent containers	6	**Straight-sided**, essential: clear plastic cups, small square-section jars, or cuvettes if your department has them. They must all be the same, because the container is part of the light path.
Coloured stock solution	250 mL	Food colouring in water is the most reliable. **Red or magenta gives the biggest signal** with this sensor: they absorb green, where the sensor is most sensitive. Blue and green absorb in the red, where the sensor sees little, and give small absorbances. Tea and coffee work but are weaker and must be filtered. Milk and orange juice do *not* work: they scatter light rather than absorb it.
Distilled water	1 L	For the dilutions and for the blank. Tap water is acceptable if distilled is unavailable, but use the same water throughout.
Syringe, 10 or 20 mL	1	For the dilution series. A syringe without a needle is safer and easier than a pipette for most classes.
Measuring cylinder, 100 mL	1	So that each dilution is made up to the same total volume.
Torch or small desk lamp	1	It must be *steady*. Mains LED lamps are ideal. Avoid battery torches whose output fades, and any lamp that flickers. **A coloured LED** of the colour the dye absorbs (green for a red or magenta dye) is better still: nearly monochromatic light straightens the line.
Cardboard box or dark cloth	1	To exclude room light. A shoebox with two holes makes an excellent light tunnel.
Ruler	1	For the internal path length l of the container, in cm, and to fix the lamp-to-sensor distance.
Marker, labels, stirring rod	1 set	Six unlabelled cups of coloured water are six unknown samples: label before filling, and mix each dilution properly.
Paper towel and a tray	1 set	Spills are certain.

Preparation the day before. Make up a single batch of stock for the whole class — for example 10 mL of red or magenta food colouring made up to 1 litre with distilled water, or four tea bags brewed in a litre, cooled and filtered through a coffee filter to remove particles. Every sample in the class is then made by dilution **from this same stock**. Label it STOCK and treat it as $c = 1.00$.

Then the step teachers most often skip: **run the experiment yourself once**, with the blank and the stock. You are checking one thing — that the stock still transmits a usable amount of light. **If it reads below about five per cent of the blank, dilute it and try again**; a solution that blocks nearly all the light gives an absorbance that is pure noise. Aim for a stock absorbance between 0.8 and 1.2. Finally, decant into labelled bottles, one per group, so that thirty pupils are not queuing at a single container.

4. Scientific background for the teacher

The law, derived from a single physical assumption

Consider a parallel beam of light of intensity I passing through an absorbing solution. Look at a thin slice of thickness $\mathrm{d}x$. The only assumption needed is this: *the fraction of the light removed by the slice is proportional to the number of absorbing molecules it contains*, that is, to the concentration c and to the thickness $\mathrm{d}x$:

$$\mathrm{d}I = -k\,c\,I\,\mathrm{d}x.$$

Note the I on the right-hand side: **a slice removes a fixed *fraction*, not a fixed amount** — precisely the condition that generates an exponential. Separating the variables and integrating over the thickness l,

$$I = I_0\,\mathrm{e}^{-k\,c\,l}.$$

The definition of absorbance,

$$A = \log_{10}\left(\frac{I_0}{I}\right),$$

turns the exponential into a straight line: $A = \varepsilon\,c\,l$, with $\varepsilon = k/\ln 10$. **That is the whole story**, and pupils should be able to reproduce it on the board: proportionality to the fraction gives the exponential, and the logarithm straightens it.
Useful landmarks: $A = 0$ means all the light passes; $A = 1$ means a tenth passes; $A = 2$, a hundredth.

4.1 What the gradient means

On a plot of A against relative concentration c, the gradient is $\varepsilon\,l$. If the path length l is the same for every sample — and it is, if the containers are identical — then the gradient measures directly how strongly the substance absorbs, under your lamp's light. Two different dyes give two different gradients; the same dye measured by two groups with different lamps also gives two different gradients, and that is a result, not an error.

Once the line is drawn, an unknown sample is determined by reading off the concentration corresponding to its measured absorbance. **This is exactly what a laboratory colorimeter does**, and the pupils have just built one.

4.2 Where the law breaks down — state this honestly

Beer–Lambert is an idealisation and every one of its assumptions can be broken in a school laboratory. Pupils are told this in advance, and the over-concentrated sample exists so that they can *see* it break.

1. **The source is not monochromatic — the main limitation here.** The law holds at one wavelength. The sensor is a **broadband** detector with a roughly eye-like response, and the lamp emits a wide spectrum. If the light contains two wavelengths with different coefficients,

$\varepsilon_1 \neq \varepsilon_2$, the detector sums them:

$$I = I_{01}10^{-\varepsilon_1 cl} + I_{02}10^{-\varepsilon_2 cl}.$$

The weakly absorbed component dies away more slowly and eventually dominates the total, so the measured absorbance **bends downwards** away from the straight line at high concentration. This negative deviation is unavoidable without a filter or a monochromator. Be honest with pupils: their instrument is a colorimeter, not a spectrophotometer, and this is the specific difference. **Slipping a complementary coloured filter in front of the sensor narrows the band and visibly improves the linearity** — an excellent extension.

2. **Stray light.** Any room light reaching the detector without passing through the sample adds a constant I_s to both readings, so the measured absorbance is

$$A_{\text{meas}} = \log_{10}\frac{I_0 + I_s}{I + I_s} \xrightarrow[I\to 0]{} \log_{10}\left(1 + \frac{I_0}{I_s}\right).$$

The absorbance therefore **saturates at a ceiling** set by the stray-light ratio, no matter how concentrated the solution becomes. This single equation explains most of the flattening pupils observe, and it is why the dark box is compulsory rather than optional.

3. **Scattering and turbidity.** Tea leaf fragments, coffee grounds, bubbles or a cloudy suspension remove light by *scattering* it out of the beam rather than absorbing it. The detector cannot tell the difference, so the apparent absorbance is too high and does not scale with concentration in the same way. Filter the stock, tap the cups to release bubbles, and do not use milk or orange juice.

4. **High concentration.** Above roughly 10^{-2} mol L^{-1} the absorbing species begin to interact: dimerisation, changes in refractive index, shifts in activity coefficients. ε is then no longer constant and the law fails on chemical grounds even with a perfect instrument.

5. **Detector behaviour.** With firmware 2.2.5 or newer, the sensor switches gain and integration time automatically over a very wide range. **Before 2.2.5 the reading stopped at about 3800 lux**, at one fixed sensitivity — and a nearby lamp easily exceeds that. Near the bottom the reading becomes coarse with either version. **Keep I_0 comfortably inside the range** — a few hundred to about 3000 lux — rather than driving the lamp as hard as possible.

6. **Reflection at the container walls.** Every air-plastic and plastic-liquid interface reflects a few per cent of the light. This is exactly why the blank must be **solvent in an identical cup**, not an empty light path: that cancels the wall reflections and leaves only the absorption due to the dye.

Technical note — I_0 is shown on screen and can be re-taken

The sensor. The light sensor in the box (the `LUX` column in the exported files).
The single most important thing in this guide. Below the cards there is a **reference row**. It states what I_0 currently is and whether it was *chosen by you* with the button — *Reference I_0 = … lx (set by you, on the blank)* —, followed by the name of the box being read. Beside it is the button **Set reference $\mathbf{I_0}$ from this reading**; it takes the mean of the last five seconds, not a single reading.
Read that row before measuring anything. The blank — the cup of water — is compulsory: put it in the beam and press the button. If the row does not say that the reference was chosen by you (it is unset, or was taken automatically by the app), I_0 cannot be trusted. If a group started with a dark solution already in the beam, put the blank in the beam and press the button; the absorbances afterwards are right again, without restarting the activity.
The classroom practice stays what it was, though, and the sheet still enforces it: **record**

the raw illuminance in lux for every sample and compute the absorbances by hand afterwards. That is no longer insurance against a bad reference — it is simply the point of the lesson: pupils own the logarithm rather than watching a card produce it.
The box. The activity header shows the box the readings come from. If several are connected, choose the group's box in the header and check it at a glance.
The cards, the table and the chart. At the calibration step, **Illuminance $\mathbf{I_0}$** appears, in lux, with the on-screen warning to *note the value of I_0 on paper*. At the three measuring steps, **Illuminance I** and **Absorbance A** appear, with the **Sample in the beam:** list holding that step's samples (from *c = 0.2 (20 mL of stock)* to *c ≈ 2 (extra dye, on purpose)*) and the **Capture** button. Each press puts the mean of the last five seconds into the chosen sample's row of the **Sample / I (lx) / T (%) / A** table and moves the list on to the next sample. The illuminance chart beside it shows how the light changed from one sample to the next.
The A against c graph. The analysis step draws the samples up to the stock (filled), the extra-dye sample (open circle) and the **fitted line** through the points with $c \leq 1$, and above it writes *Slope b = … ± … · intercept a = … · R^2 = … (n points, c ≤ 1)*. The "±" is the standard error of the slope; with only two points it is missing, because there is no scatter to compute it from.
The blank again. At the **How sure are we? The blank again, and the scatter** step, the blank goes back in the beam and **Capture** is pressed; the app writes *Blank at the end: … lx, … % away from I_0*. Above 10% it adds that the lamp, the cup or the box moved and that the series must be measured again.
The steps and their buttons. The green button changes its label at every step: **I understand the theory!**, **My prediction is written**, **$\mathbf{I_0}$ calibrated**, **Dilute solution noted**, **Medium solution noted**, **All concentrations noted**, **Graph plotted**, **Blank checked** and **Finish**. If a measuring step ends with nothing captured, the button turns pale and a hint asks for a steady reading; you can still move on, but the row stays empty.
Exporting the data. At the measurement steps, **PNG** and **Excel** appear; Excel saves the step's raw lux, unfiltered, with an `elapsed_s` column, plus the group's journal. **There is no CSV button inside the activity**; the CSV file is taken from the **Data & Export** page. **The raw data are in lux**: the absorbance is computed from them and from I_0 — one more reason to write I_0 down on paper.

5. Lesson flow

The in-app wizard has nine steps: the theory, a written prediction, calibration of I_0 on the blank, three measuring steps (the dilute solutions $c = 0.2$ and 0.4; the medium ones, 0.6 and 0.8; the stock and the extra-dye sample), graphical analysis, **How sure are we?** and the conclusion with the applications. Below is what the teacher does around them.

1. **Elicit written predictions first (5 min).** Before any equipment is switched on, every pupil answers in writing: *if doubling the concentration halves the transmitted light, what will tripling it do? Sketch the graph of transmitted light against concentration. Now sketch the graph of $\log_{10}(I_0/I)$ against concentration.* Most classes draw the first graph as a straight line falling to zero. **Keep those sketches:** comparing them with the data at the end is the sharpest moment of the activity. The group's prediction is also typed into the app's prediction step, which quotes it in the conclusion.

2. **Theory (8 min).** Derive $I = I_0 e^{-kcl}$ on the board from the thin-slice argument, emphasising the physical reason for the exponential: each slice removes a fixed fraction. Then define absorbance and show that the definition is exactly what makes the plot straight.

3. **Build and align the rig (8 min).** Lamp at one end of the box, sensor at the other, a

marked position for the cup between them. Measure and record the lamp-to-sensor distance and the internal width l of the cup. Tape everything down. **Nothing may move again.**

4. **Prepare the dilution series (15 min, lesson 1).** With the syringe and the measuring cylinder, each group makes five samples up to the same total volume — 20, 40, 60, 80 and 100 mL of stock made up to 100 mL — plus pure water as the blank. All of them come **from the same stock**, not one from another. Mind the wording: a "1:2" dilution (one part solution, one part water) *halves* the concentration, it does not double it. Label before filling.

5. **Measurement — the blank first (lesson 2). The first sample in the beam is the blank**: with it in the beam press **Set reference I_0 from this reading**, and the reference row should then say that the reference was chosen by you. Record the value from the card *on paper*. Then each concentration in increasing order: choose the sample in the **Sample in the beam:** list, wait five seconds and press **Capture**. The app's table keeps one value per sample (the last one captured); the three readings the sheet asks for are written on paper, each time in **lux**, not just the absorbance on screen. The over-concentrated sample is last.

6. **Graphical analysis (lesson 3).** The app's analysis step shows the A against c graph at once, with the fitted line, the slope with its "±" and R^2 — use it as a check, not as the result. Pupils compute T and A from the lux themselves, plot A against c, fit a line to the linear part and compute the gradient. Pupils identify for themselves *where* the points leave the line — with white light, at high concentration the points fall below it because the light is polychromatic — and state the range over which their instrument is trustworthy.

7. **How sure are we? (3 min).** The blank goes back in the beam and is captured. If the app shows it more than a few per cent away from I_0, the lamp has drifted; above 10% the series is no longer comparable. The step also points out that the "±" beside the slope comes from the scatter of the points about the line, and that three readings per sample and a second series say more than a single run.

8. **Applications and conclusion.** The app's final step puts side by side the group's prediction, I_0, the slope with its "±", R^2, the extra-dye sample's absorbance against what the line predicts and the blank at the end, then lists where the law is used: analytical chemistry, pulse oximetry, stellar atmospheres, quality control, water turbidity. Close by asking every pupil to compare the measured graph with the sketch they drew at the start of the lesson.

The investigation, in one table

The question. Is absorbance proportional to concentration — and up to what concentration?
The hypothesis. Written **before** any measuring, as two graph sketches: transmitted light against concentration, and $\log_{10}(I_0/I)$ against concentration. Not changed afterwards.
What I change. One thing only: the **concentration** of the solution.
What I measure. The illuminance, in lux — not the absorbance on the screen. I compute the absorbance myself, from the lux and from the I_0 written on paper.
What I keep the same. The same lamp at the same distance; identical containers filled to the same depth; the same marked position in the box; the box closed every time; and **the same I_0**, captured on the blank at the start of the session.
How many repeats. At least three readings per concentration, and a return to the blank at the end. **If the final blank no longer gives nearly the same value as I_0, the lamp has drifted** and the series is no longer comparable.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the box closed, the cup wiped and free of bubbles, I_0 captured on the blank and the absorbance below 1. An artefact is anything else: **the worst possible mistake is for I_0 to have been captured through a coloured solution** — then every absorbance

comes out close to zero until the blank goes back in the beam and **Set reference I_0 from this reading** is pressed again. Stray light flattens the top end; a bubble or a fingerprint changes the transmission by more than a dilution step does.

6. Analysis after the activity

6.1 What the data should look like

Sample	Transmittance	Absorbance	Comment
Blank (water in a cup)	100%	0.00	By definition; this is where I_0 is captured.
$c = 0.2$	70–85%	0.07–0.15	The bottom of the line.
$c = 0.4$	50–70%	0.15–0.30	
$c = 0.6$	35–55%	0.26–0.46	
$c = 0.8$	25–45%	0.35–0.60	
$c = 1.0$ (the stock)	10–25%	0.6–1.0	The target set during preparation.
Over-concentrated sample	under 5%	above 1.3, but *below* what the line predicts	**This is where the law is seen to break.**

Absolute values depend on the dye, the lamp and the geometry. What must come out reliably is **the shape**: a good straight line through the origin at the bottom, then a downward bend at the top.

6.2 The graphical analysis, step by step

1. **From lux to transmittance.** $T = I/I_0$, as a percentage. It is a simple division, and it is worth doing before the logarithm so that pupils see the curve.
2. **From transmittance to absorbance.** $A = -\log_{10} T$, with T as a fraction, not a percentage — **this is the commonest arithmetic mistake**. Quick check: $T = 10\%$ must give exactly $A = 1$.
3. **The graph.** A on the vertical axis, c on the horizontal. Draw the line **through the linear points only**, not through all of them, and extend it through the origin — the line *must* pass through the origin, because at zero concentration nothing is absorbed.
4. **The gradient and R^2.** The gradient is εl. Compute R^2 on the linear points and report it *together with the number of points used* — an excellent R^2 on three points means nothing.
5. **Stating the range.** The last step, and the most valuable: *our instrument is linear up to $A \approx$ ____, and above that it underestimates the concentration.* That is an analyst's statement, not a pupil's.

6.3 Questions to ask afterwards

1. Compare the measured graph with the sketch you drew at the start. What did you draw wrongly, and why was that intuition natural?
2. Your line bent downwards at high concentration. Give two distinct explanations — one about the solution, one about the instrument — and say how you would tell them apart experimentally.

3. Why must the blank be water *in a cup* rather than an empty light path? What exactly does that cancel?
4. If you doubled the width of the cup instead of doubling the concentration, what would happen to the absorbance? Justify it with the formula.
5. What would have happened to all your results if I_0 had been taken with the concentrated solution in the beam instead of the blank, and how would you have spotted it?

6.4 Assessment ideas

- **Low stakes:** the pupil completes the table of lux values and computes the transmittance correctly for each sample, then ranks them.
- **Standard:** a full table with T and A, the A against c graph with a line of best fit, the gradient computed, and a sentence saying where the points leave the line.
- **Extension:** determining an unknown by interpolation with a stated uncertainty; repeating the series through a complementary coloured filter and comparing the linearity of the two; or varying the path length instead of the concentration, showing that A depends on both in the same way.

Common pitfalls and misconceptions

I_0 captured through a solution. Still the easiest mistake to make. It is repaired on the spot: blank in the beam, **Set reference $\mathbf{I_0}$ from this reading**, done. **Read the reference row before the first measurement** — it is the only check needed.
Taking the logarithm of a percentage. $A = -\log_{10} T$ requires T as a fraction. A pupil who enters 10 instead of 0.1 gets a negative number and concludes the experiment failed. The check: $T = 10\%$ gives $A = 1$.
The blank as an empty light path. That cancels the solvent's absorption but not the reflections at the cup walls. The blank is water in an identical cup.
Room light. It puts a ceiling on the absorbance, however concentrated the solution, and it is the main cause of the flattening. The dark box is not optional.
A drifting lamp. A battery torch fades measurably over twenty minutes, and I_0 was captured at the start: every later absorbance reads high. Measure the blank again at the end.
Bubbles and particles. They scatter light, so the absorbance looks larger but does not scale with concentration. Filter, tap the cup, do not use milk or orange juice.
A blue or green dye. It absorbs in the red, where the sensor is not very sensitive, so the absorbances come out small and the line bends early. Choose red or magenta, or light the rig with a coloured LED.
Dilutions made one from another, or "1:2" read as doubling. Chained dilutions add up their errors; every sample is made from the same stock. A 1:2 dilution halves the concentration.
A line drawn through every point. That includes the non-linear region and ruins both the gradient and the conclusion. Draw through the linear part only, and discuss the rest.
Moving anything. The lamp, the cup, the sensor, the box. **A 10% change in distance changes the intensity by about** 20%, which is more than a dilution step.

Safety and classroom management

Dyes and solutions. Food colouring and tea are low hazard but stain skin, clothing and benches — gloves, aprons and immediate mopping of spills. If you use ink, check its safety data. **Nothing is tasted**, including the tea and coffee used as samples.
Eye protection is worn when pouring and diluting.

Lamps get hot, particularly filament ones. Keep them clear of card and paper — **the light tunnel is itself a cardboard box**, so the lamp is fixed at the hole, not inside it, and is never covered. Let the lamp cool before packing away.
Liquids and electronics. The box is not waterproof: it and the laptop stay on a different part of the bench from the pouring area, on a tray. Fill the cups away from the rig, wipe them, and only then place them in the beam.
Glassware. Prefer plastic cups. If glass is used, check for chips.
Syringes are always without needles.

Adaptations for pupils with special educational needs

The logarithm column, pre-computed — but only halfway. This adaptation needs care, because *the logarithm is the lesson*. Do not give the A column ready-computed. Give instead, printed on the sheet, a **fully worked example** for the first row — lux, division, logarithm, result — which the pupil copies as a pattern for the rest. The support is in the model, not in the answer.
The dilution table, pre-filled. The volumes of stock and water for each concentration arrive printed. The arithmetic of dilution is not the skill being assessed here.
Axes drawn in advance. A sheet with the axes drawn, scaled and labelled, A from 0 to 1.5 and c from 0 to 1. The pupil places the points and draws the line. The idea being learned is *where the points leave the line*, not the drawing of axes.
Three real roles for pupils with motor difficulties. Not being able to handle a syringe does not mean not taking part. The roles rotate at every sample:

- **the blank keeper** — makes sure the **first** sample in the beam is the blank and that I_0 is written on paper before anyone moves on; this is the role with the greatest scientific value in the group, because a mistake here voids the whole lesson;
- **the screen reader** — says the illuminance in lux aloud and announces if the value will not settle;
- **the scribe** — writes the lux into the table and computes T as the readings arrive, which makes a bad sample visible on the spot.

Sensory sensitivity. The dark box and laboratory smells can be unpleasant. The scribe and screen reader roles are done at a distance.

If things do not go to plan

Every absorbance comes out close to zero. Almost certainly I_0 was captured through a coloured solution — the reference row shows either that it was taken automatically from the first reading, or a value far below the blank's. **Put the blank in the beam and press Set reference $\mathbf{I_0}$ from this reading**; the measurements afterwards are correct. And if you recorded the raw lux — and the sheet requires it — the earlier series **can be saved** too: recompute everything by hand, using the blank's value as I_0.
The stock blocks nearly all the light. Dilute it and start again. An absorbance above 1.5 is not a better result, it is noise.
You have no identical containers. Use *the same* cup for the whole series, rinsed and wiped between samples, in increasing order of concentration. That is actually better than "identical" cups that are not.
You have no dark box. Switch the lights off, close the blinds, lay a black cloth over the rig. If even that is impossible, measure the background with the lamp off and subtract it from every reading before computing T.
You have no time for three lessons. Run the 50-minute route with solutions you prepared. **Do not cut the analysis** to make room for the dilution: you can do the diluting, you cannot do the logarithm for them.

The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils compute T and A for each row, plot the graph, fit the line through the linear part and show where the last sample leaves it. **All the mathematics of the lesson is unchanged** — and it is the lesson. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Royal Society of Chemistry** — `https://www.rsc.org` — search for the Beer–Lambert law and colorimetry in their schools material; they have classroom-tested versions with easily obtained substances.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the Beer's law simulation; it lets pupils change concentration, path length and wavelength and watch the absorbance respond, ideally just before they measure it.
- **Institute of Physics** — `https://www.iop.org` — search for material on the absorption and attenuation of light, for the physics half of the derivation.
- **NIST** — `https://www.nist.gov` — search for spectrophotometry standards, if a pupil asks how an instrument of this kind is really calibrated.
- **Nuffield Foundation** — `https://www.nuffieldfoundation.org` — classic, carefully trialled practical work, for the low-tech versions.
- **Royal Society of Biology** — `https://www.rsb.org.uk` — for pulse oximetry and pigment absorption, if you go on to the biological applications the app lists.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *why does the logarithm straighten the graph?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Scientist level the three axes look like this:

- **Measurement practice** — the pupil puts the blank in the beam first, writes I_0 down on paper, records the raw lux for every sample, and checks the blank again at the end.
- **Data interpretation** — the pupil computes A from T as a fraction correctly, draws the line through *only* the linear part, and states the instrument's trustworthy range.
- **Scientific reasoning** — the pupil explains why the thin slice leads to an exponential, why the logarithm straightens it, and gives at least one instrumental and one chemical cause for the bend at the top.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/beer_lambert_law.pdf`. It has two pages — the two prediction sketches, the rig and the I_0 box on the first;

the table of the series, the worked example and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The sheet asks for the raw lux, not the absorbance from the screen**, and this is its central decision: the displayed absorbance depends on an I_0 that can be wrong without anybody noticing, whereas from the lux the series can always be rebuilt. **The two graph sketches are drawn at the start**, side by side, with room beside them for the real curve — comparing them is the best moment of the lesson and has to be on the same page. **The first row of the table comes with a fully worked example**, from lux to absorbance, so that the pupil has a pattern to follow; the absorbance column is *not* pre-computed, because the logarithm is the lesson. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.